

Smarter technology for all

Servicing the ThinkSystem SR680a V3

ES72657

April 2025

Lenovo

Prerequisites

- [ES42373B – Intel Xeon processor architecture for ThinkSystem V3 servers](#)
- [ES51757B – Introducing ThinkSystem tools](#)
- [ES52374 – ThinkSystem tools for the ThinkSystem V3 platform](#)
- [ES41759C – ThinkSystem problem determination](#)
- [ES51780C – Servicing the ThinkSystem storage controllers](#)

Objectives

After completing the course, you will be able to:

- Describe the ThinkSystem SR680a V3 and its components
- List the SR680a V3 specifications
- Describe the SR680a V3 configurations and block diagrams
- Describe the SR680a V3 management tools
- Describe the problem determination steps and explain how to troubleshoot issues with the SR680a V3

Product overview

Product description and front, rear, and inside views

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Product overview

The ThinkSystem SR680a V3 (machine type:7DHE) is an 8U, air-cooled, two-socket, GPU-rich rack server. The SR680a V3 is designed for demanding AI computing or HPC workloads.

The SR680a V3 supports two 4th or 5th Generation Intel Xeon Scalable processors and eight NVIDIA or AMD OAM (open project computing accelerator module) GPUs. Supported GPUs include the NVIDIA H100 and H200, and the AMD MI300X (planned).



Specifications

Features	SR680a V3
Form factor	8U rack server
Processor	Two 4 th or 5 th Gen Intel Xeon Scalable processors, TDP up to 350 W
Memory	32 DIMM slots with two processors (16 DIMM slots per processor) Each processor has eight memory channels, with two DIMMs per channel (DPC) Lenovo TruDDR5 RDIMMs are supported DIMMs operate at up to 5600 MHz at 1 DPC and up to 4400 MHz at 2 DPC
Disk drive bays	Up to 16 2.5-inch hot-swap drive bays, supporting PCIe 5.0 NVMe drives
OS boot drive	Support for two M.2 drives with optional Intel VROC NVMe RAID support for OS boot and data storage functions
PCIe slots	• Front: Eight PCIe 5.0 x16 FHHL slots with GPU direct support • Rear: Two PCIe 5.0 x16 FHHL slots
PSUs	Eight hot-swap 2600 W CFFv4 PSUs, support for N+N redundancy
Cooling	• Five hot-swap fans at the front for the compute shuttle

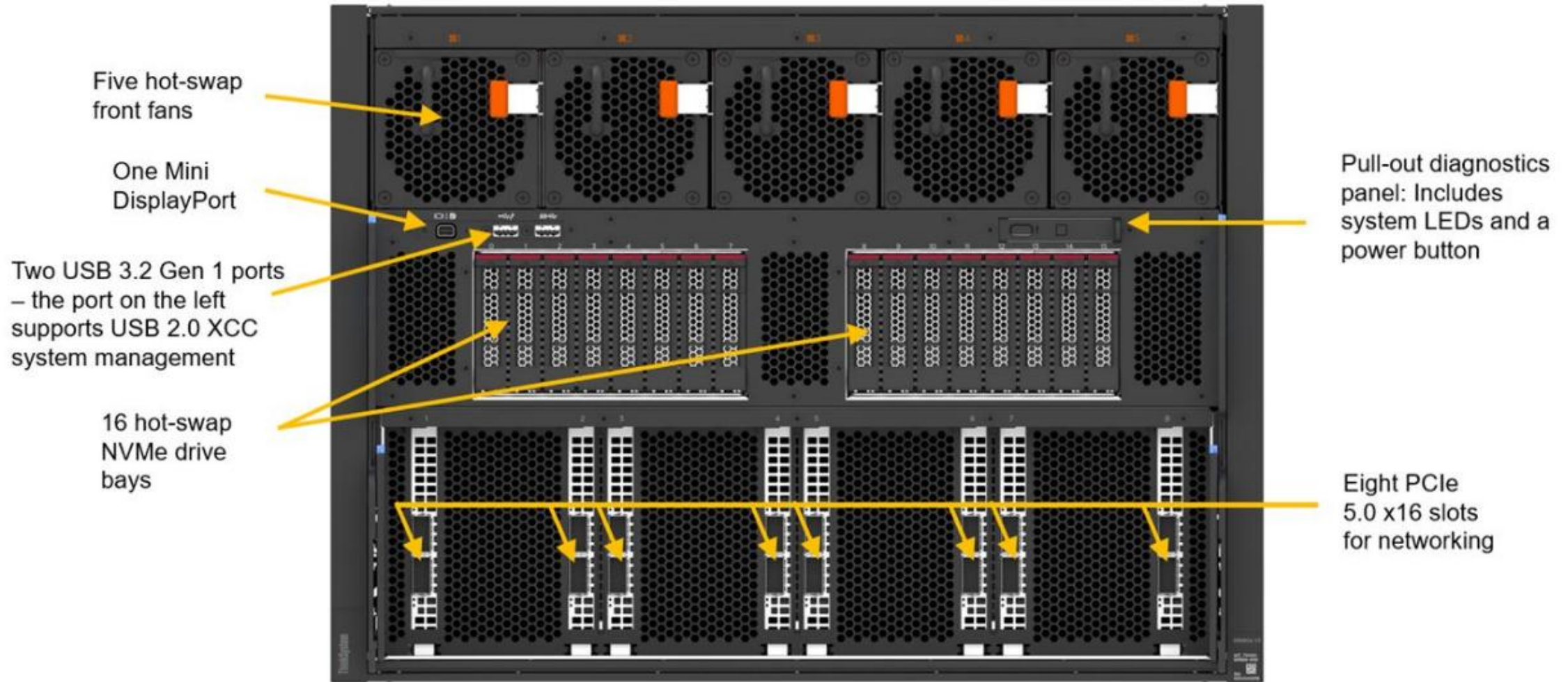


Note: For the latest specifications, refer to the product guide on [Lenovo Press](#).

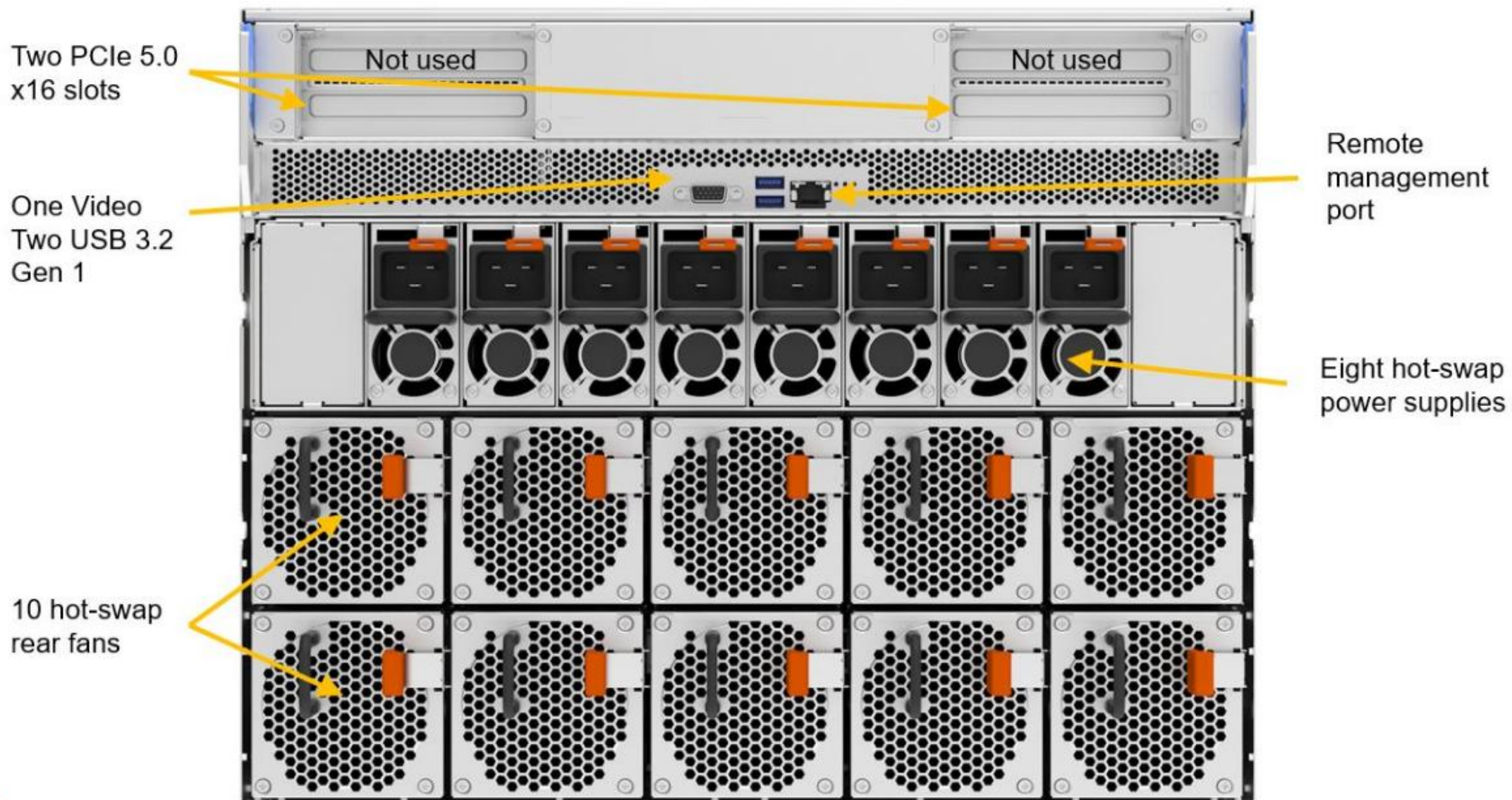
Specifications

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PSUs	Eight hot-swap 2600 W CFFv4 PSUs, support for N+N redundancy
Cooling	• Five hot-swap fans at the front for the compute shuttle • 10 hot-swap fans at the rear for the GPU shuttle • Support for N+1 redundancy across all 15 fans
GPUs	• Eight NVIDIA H100 700W SXM5 GPUs with 80GB HBM3 memory • Eight NVIDIA H200 700W SXM5 GPUs with 141GB HBM3 memory [No Title]
Management tools	XClarity series tools: • XClarity Controller 2 (XCC2) • XClarity Provisioning Manager (LXPM) • XClarity Essentials (LXCE)

Front view



Rear view



Inside views

Click the buttons to see inside views of the SR680a V3.



Exploded view

Inside view without
chassis

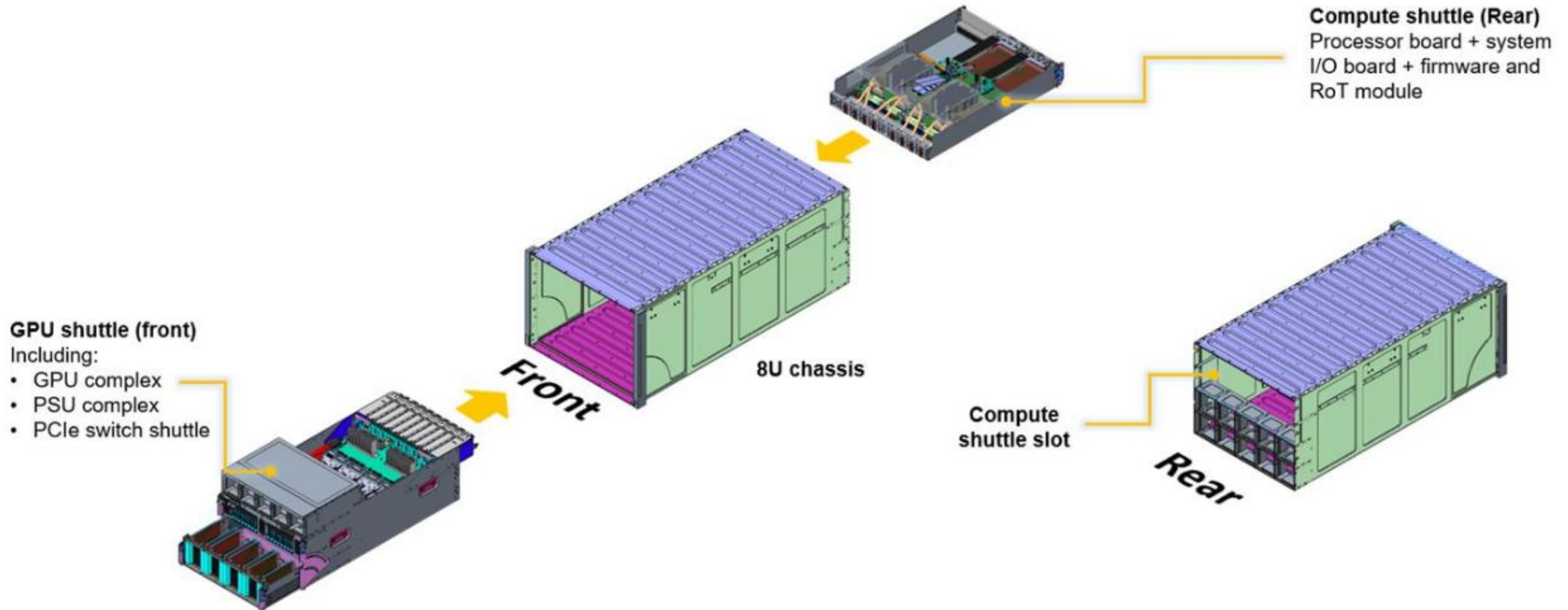
Side view

Compute shuttle

GPU shuttle

Inside views

Click the buttons to see inside views of the SR680a V3.



Exploded view

Inside view without chassis

Side view

Compute shuttle

GPU shuttle

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Front fans

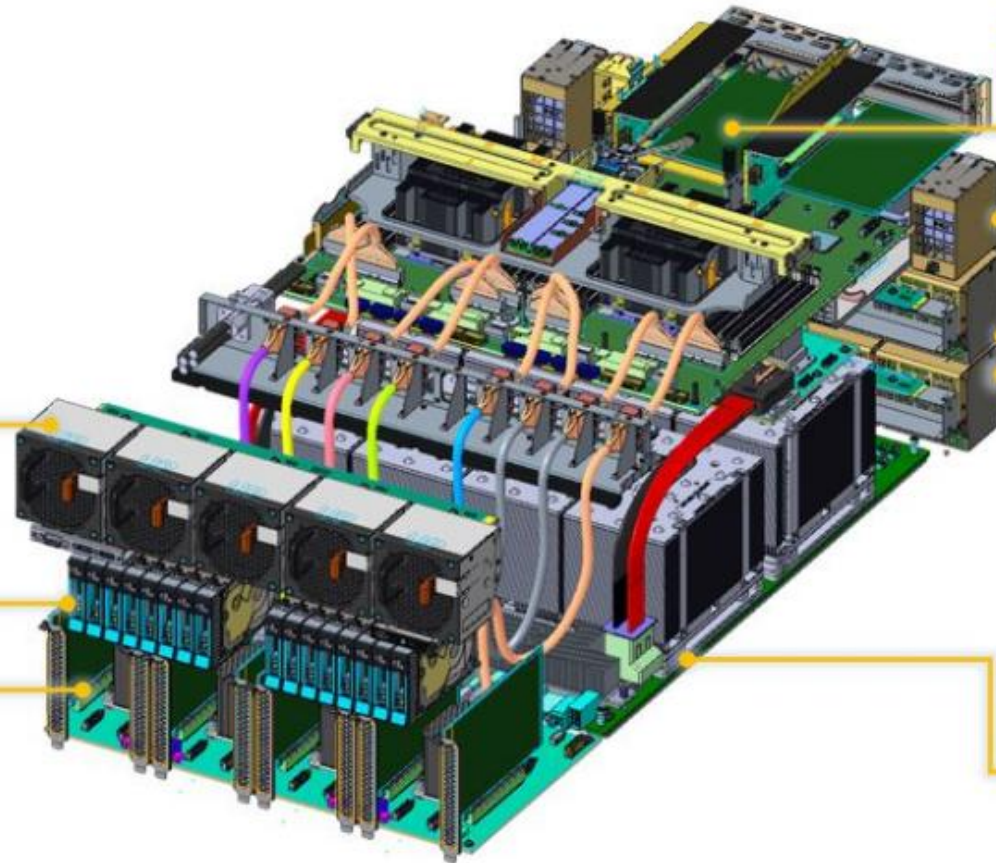
Five front hot-swap fans

Hot-swap drives

16 2.5-inch NVMe drives

Switch board

PCIe lane signal connections for internal GPUs to the system board assembly



System board assembly

Processor board + system I/O board + firmware and RoT module

PSU complex

Eight hot-swap PSUs

Rear fans

10 rear hot-swap fans

GPU complex

Includes eight NVIDIA or AMD OAM GPUs, and a GPU baseboard

Exploded view

Inside view without chassis

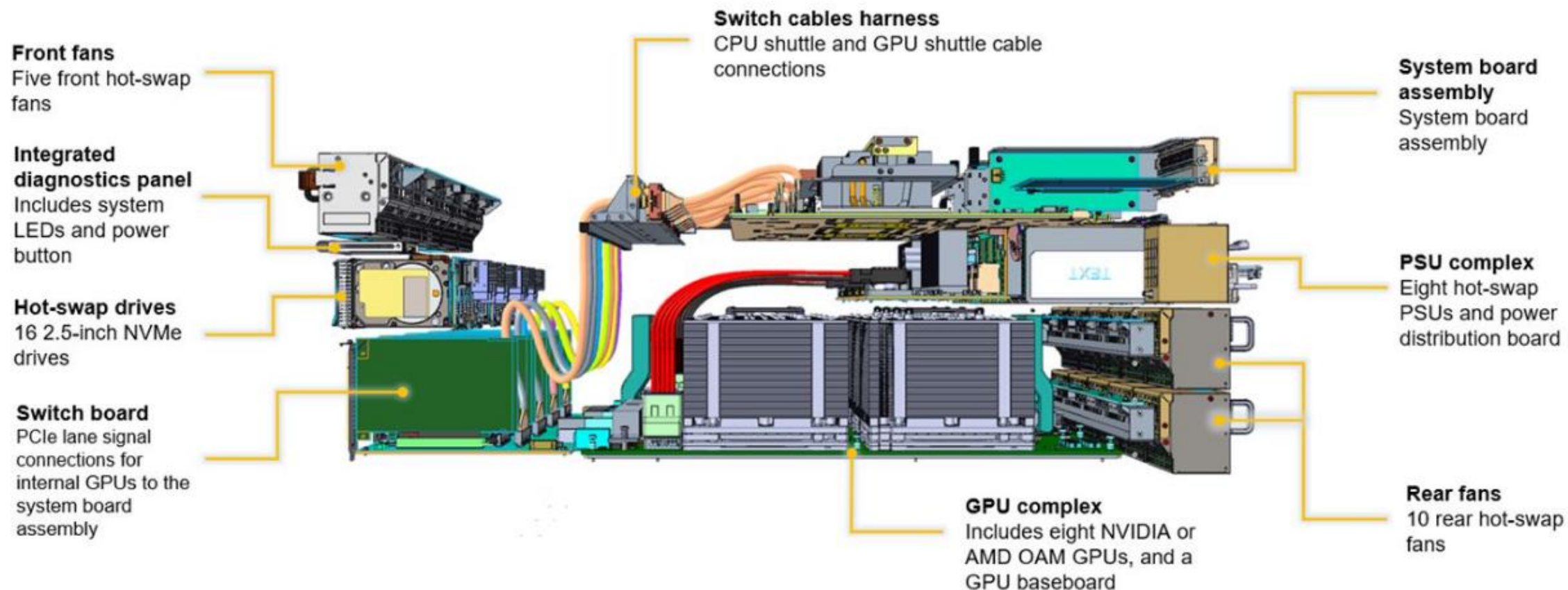
Side view

Compute shuttle

GPU shuttle

Inside views

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Exploded view

Inside view without chassis

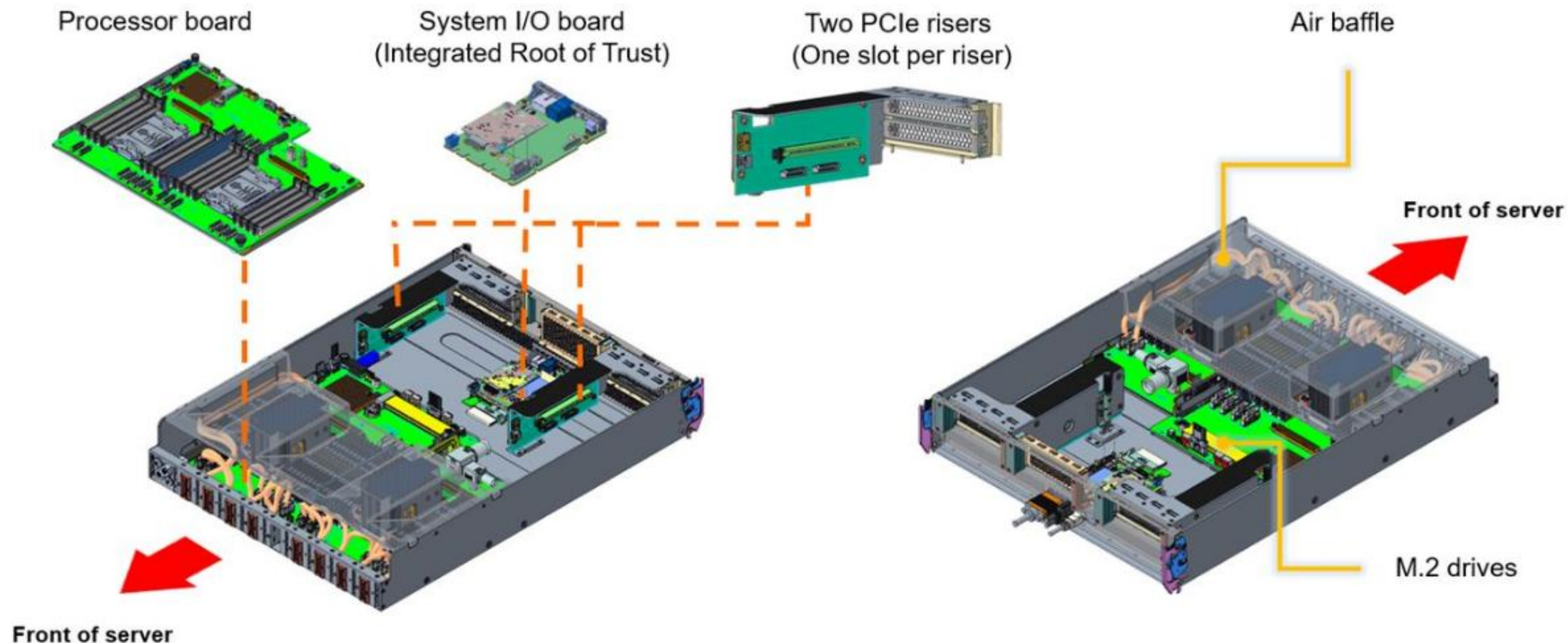
Side view

Compute shuttle

GPU shuttle

Inside views

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Exploded view

Inside view without
chassis

Side view

Compute shuttle

GPU shuttle

Inside views

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System front I/O tray

Includes a VGA port and two USB ports

Front fan board

For five front fans

PSU complex

Includes eight hot-swap PSUs and power distribution board

Two rear fan boards

For five front fans each

Switch board

PCIe lane signal connections for internal GPUs to the system board assembly

AMD OAM GPUs

MI300x

NVIDIA OAM GPUs

H100 or H200

OR

8U shuttle chassis

Exploded view

Inside view without chassis

Side view

Compute shuttle

GPU shuttle